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The Al–R–Mg (R = Gd, Dy, Ho) systems. Part II: Thermodynamic modelling of the binary and ternary systems

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Abstract

The thermodynamic modelling and optimisation of the Al–R, R–Mg and Al–R–Mg (R = Gd, Dy, Ho) systems has been carried out, the Al–Mg system being already optimised in literature. The Compound Energy Formalism (CEF) has been used to describe the thermodynamic functions of either solutions and stoichiometric phases present in the systems. In particular, the order/disorder relation between cI2-W (A2) and cP2-CsCl (B2) phases present in the systems has been taken into account and thermodynamically modelled. Moreover γ-(Mg,Al) and $R_5(Mg_{1-x}Al_x)_{24}$ (R = Dy, Ho) phases, related to the cI58-αMn (A12) type structure, have been modelled as different sublattice occupations of the same A12 phase. The thermodynamic modelling of the ternary systems is mainly based on the experimental investigation carried out in our laboratory and reported in a separate paper in this issue [1]. As a result, a complete description of the solid-solid and solid-liquid phase equilibria in the six binary and three ternary systems has been obtained.
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1. Introduction

In this paper the results obtained in the thermodynamic modelling and optimisation of six binary and three ternary systems are presented. The systems studied are Al–Gd, Al–Dy, Al–Ho, Gd–Mg, Dy–Mg, Ho–Mg, Al–Gd–Mg, Al–Dy–Mg and Al–Ho–Mg. In the following, for brevity, Gd, Dy and Ho will be denoted by R; Ln however will be used to generally denote trivalent lanthanide elements (from La to Lu except Eu and Yb).

The combination of experiments and thermodynamic calculations is a powerful strategy when investigating phase equilibria in multi-component systems. Indeed it was successfully applied to the study of the Al–Sc–Mg [2–4] Al–Gd–Mg [5] and Al–Er–Mg [6,7] systems.

Also in this work experiments (presented in part I, a separate paper in this issue [1]) and thermodynamic calculations were carried out by an iterative procedure: calculations were used to interpret experimental results and to select new key experiments; experimental results, were used to refine the thermodynamic description of the phases and to obtain more reliable calculations.

Special attention was also given to the systematics of the rare earth behaviour. It is well known indeed that phase equilibria, thermodynamics and, in general, constitutional properties of the rare earth alloys show smooth and regular variations when passing from one element to the next along the lanthanidic series. In the present investigation three heavy rare earths are involved (Gd, Dy and Ho) and their binary and ternary alloys with Al and Mg are expected to show similar behaviour and properties.

2. Experimental information

The experimental results available in literature on phase equilibria in the binary and ternary Al–R, R–Mg and Al–R–Mg systems considered in this work have already been described in part I. In the following they are briefly summarised together with data concerning thermodynamics.

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2.1. Al–R systems (R = Gd, Dy, Ho)

The room temperature enthalpy of formation of the $LnAl_2$ phases has been measured for all the lanthanide elements by Colinet and Pasturel [8] by aluminium solution calorimetry.

A critical assessment of the Al–Gd phase diagram has been reported by Gschneidner and Calderwood [9]. More recently this system has been investigated by Saccone et al. in the composition range 0–66.7 at.% Al [10]. The enthalpy of formation of the solid alloys at 298 K has been measured by Al solution calorimetry by Colinet et al. [11] for all the intermediate compounds and by Sommer et al. [12] for $GdAl_2$, GdAl and Gd_2Al. The partial enthalpy of mixing of Gd at infinite dilution in liquid aluminium has been determined at 1198 K by Lee and Sommer [13]. Kober et al. [14] obtained the Gd chemical potential in the solid alloys at 800 K from e.m.f. measurements. A thermodynamic optimisation of the Al–Gd system has been recently performed by Gröbner et al. [5].

The investigation of the Al–Dy phase diagram, partially assessed by Gschneidner [15] in the composition range 66.7–100 at.% Al, has been completed by Saccone et al. [10] in the composition range 0–66.7 at.%Al.

The Al–Ho phase diagram has been experimentally determined by Meyer [16] and subsequently assessed by Gschneidner [17].

2.2. R–Mg systems (R = Gd, Dy, Ho)

Systematic measurements of the Mg vapor pressure in the solid alloys at high temperature have been carried out by Ogren et al. [18] in the Ln + LnMg two-phase field and by Pahlman and Smith [19] in the whole composition range. Ogren et al. [18] determined the Mg chemical potential and the integral Gibbs energies of formation of the LnMg (Ln = La to Lu except Eu and Yb) phases taking into account the Mg solubility in the different Ln determined by [20]. Results from [18] show a smooth and regular trend as a function of the Ln atomic number, which is in good agreement with the corresponding trend of the volume contraction. But Pahlman and Smith [19] obtained less regular trends for calculated partial and integral quantities of all the intermediate phases in the Ln–Mg systems with Ln = Ce, Nd, Gd, Dy, Er and Lu. Recently Hu et al. [21] employed an analytic modified embedded atom method (MEAM) to calculate thermodynamic properties in the Ln–Mg (Ln = Sc, Y, Pr, Nd, Gd, Tb, Dy, Ho, Er) systems. In particular dilute-limit heats of solution and enthalpies of formation of disordered solid solutions and ordered intermetallic compounds have been calculated.

The solid solubility limit of Mg has been measured by Joseph et al. [20] for La, Nd, Gd, Dy and Lu.

Phase equilibria in the Gd–Mg system have been investigated by Manfrinetti et al. [22]. Solid solubility

Table 1
Summary of the sublattice models adopted for the solution phases (predominant element in each sublattice, if any, is underlined). Concise crystallographic information relevant to the modeling has been included. Phase labels are used for phase identification in the figures

Phase label	Phase identification according to the structure	Phase identification according to the composition	Ideal occupation of the Wyckoff positions in the prototype structure and coordination (AET code)	Sublattice model (R = Gd,Dy,Ho)	Comments
	Liquid			(Al,R,Mg)	Disordered solution
01	cF4-Cu/A1	Al	Cu in 4a (AET 12-b)	($\underline{Al}$,R,Mg)	Disordered solution
02	cI2-W/A2	(β-Gd), (β-Dy), β-(Ho,Mg)	W in 2a (AET 14-b)	(Al,$\underline{R}$,Mg)	Disordered phase related to B2
03	hP2-Mg/A3	(α-Gd), (α-Dy), (Ho), (Mg)	Mg in 2c (AET 12-d)	(Al,$\underline{R}$,$\underline{Mg}$)	Disordered solution
04	cP2-CsCl/B2	$R(Mg_{1-x}Al_x)$ or RMg	Cl in 1a (AET 14-b) Cs in 1b (AET 14-b)	$(Al,\underline{R},\underline{Mg})_{0.5}(Al,R,\underline{Mg})_{0.5}$	Ordered phase related to A2
05	cI58-αMn/A12	γ-(Al,Mg) $R_5(Mg_{1-x}Al_x)_{24}$ or R_5Mg_{24}	Mn in 2a (AET 16-a) Mn in 8c (AET 16-a) Mn in 24g' (AET 12-a) Mn in 24g'' (AET 13-a)	$(R,\underline{Mg})_5(Al,\underline{Mg})_{12}(Al,\underline{Mg})_{12}$ Or $(\underline{R},Mg)_5(Al,\underline{Mg})_{12}(\underline{Al},Mg)_{12}$ (16-a) (13-a) (12-a)	The two binary phases correspond to different occupation states of the same sublattices
06	cF24-$MgCu_2$/C15	$R(Al_{1-x}Mg_x)_2$ or RAl_2	Mg in 8a (AET 16-a) Cu in 16d (AET 12-a)	$(Al,\underline{R},Mg)_1(\underline{Al},R,Mg)_2$ (16-a) (12-a)	Laves phase
07	hP12-$MgZn_2$/C14	$R(Mg_{1-x}Al_x)_2$ or RMg_2	Zn in 2a (AET 12-a) Mg in 4f (AET 16-a) Zn in 6h (AET 12-a)	$(Al,\underline{R},Mg)_1(Al,R,\underline{Mg})_2$ (16-a) (12-a)	Laves phase
08	hP24-$MgNi_2$/C36	$(Mg_{1-x}R_x)Al_2$	Mg in 4e (AET 16-a) Mg in 4f' (AET 16-a) Ni in 4f'' (AET 12-a) Ni in 6g (AET 12-a) Ni in 6h (AET 12-a)	$(Al,\underline{R},\underline{Mg})_1(\underline{Al},R,Mg)_2$ (16-a) (12-a)	Laves phase

limit of Gd in Mg has been determined by Rokhlin et al. [23,24]. The enthalpy of formation at 300 K of the GdMg phase has been measured in our laboratory [25] by direct drop calorimetry. A thermodynamic optimisation of the Gd-Mg phase diagram has been performed by Cacciamani et al. [26].

The Dy–Mg phase diagram in the whole composition range has been studied by Saccone et al. [27]. Solid solubility limit of Gd in Mg has been determined by Rokhlin et al. [28].

The Ho-Mg phase equilibria have been experimentally determined by Saccone et al. [29]; the (Mg) solvus curve and Mg-rich equilibria have been studied by Rokhlin et al. [30].

2.3. Al–Gd–Mg, Al–Dy–Mg and Al–Ho–Mg ternary systems

Very few experimental data were available in literature on these systems before the investigation carried out in our lab and reported in part I. Rokhlin et al. [31, 32] partially investigated the Al–Gd–Mg system and, on the same system, Gröbner et al. [5] performed key experiments and thermodynamic calculations. Their results, described in detail in part I, have been taken into account in the present optimization. No experimental information is available in literature for the Al–Dy–Mg and Al–Ho–Mg systems.

3. Thermodynamic models

Phases present in the optimised systems have been modelled according to the Compound Energy Formalism (CEF, [33]) and sublattice models here adopted for each phase are summarised in Tables 1 and 4. Detailed

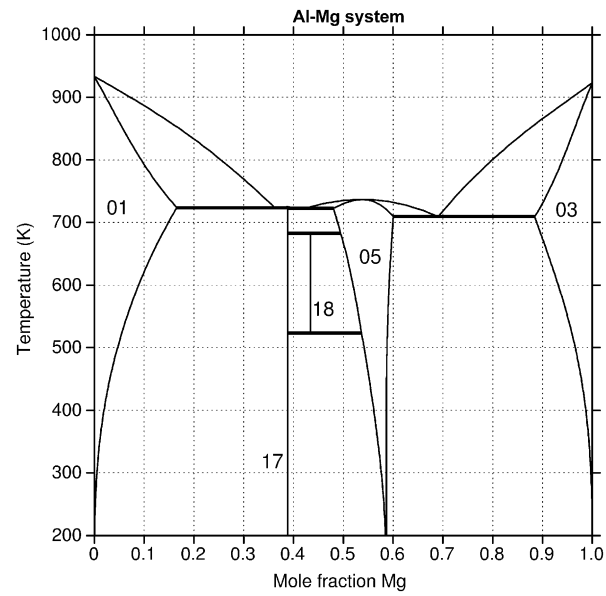

Fig. 1. The computed Al–Mg phase diagram [40]. Phases are identified by two-digit labels referring to Tables 1 and 4.

Table 2
Comparison between experimental and computed invariant equilibria in the optimised R-Al systems

Reaction	T(K) experimental	T(K) calculated	x(Al, liquid) experimental	x(Al, liquid) calculated
Gd–Al system	Refs. [9,10]		Refs. [9,10]	
(β-Gd) $\rightleftarrows$ L + (α-Gd)	1473	1471	0.095	0.087
L $\rightleftarrows$ (α-Gd) + Gd_2Al	1138	1141	0.230	0.230
L + Gd_3Al_2 $\rightleftarrows$ Gd_2Al	1213	1214	0.280	0.317
L + GdAl $\rightleftarrows$ Gd_3Al_2	1243	1268	0.310	0.394
L + Gd_2 Al $\rightleftarrows$ GdAl	1343	1341	0.440	0.443
L $\rightleftarrows$ $GdAl_2$	1793	1782	0.667	0.667
L + $GdAl_2$ $\rightleftarrows$ $GdAl_3$	1398	1403	0.840	0.860
L $\rightleftarrows$ $GdAl_3$ + Al	923	903	0.950	0.968
Dy–Al system	Refs. [10,15]		Refs. [10,15]	
(β-Dy) $\rightleftarrows$ L + (α-Dy)	1573	1571	0.100	0.04
L $\rightleftarrows$ (α-Dy) + Dy_2Al	1213	1200	0.220	0.236
L + Dy_3Al_2 $\rightleftarrows$ DyAl	1263	1263	0.270	0.320
L + DyAl $\rightleftarrows$ Dy_3Al_2	1298	1304	0.325	0.370
L + Dy_2Al $\rightleftarrows$ DyAl	1383	1396	0.450	0.430
L $\rightleftarrows$ $DyAl_2$	1773	1766	0.667	0.667
L + $DyAl_2$ $\rightleftarrows$ $DyAl_3$(ht)	1363	1364	0.886	0.888
L $\rightleftarrows$ $DyAl_3$(rt) + Al	909	916	0.975	0.979
Ho–Al system	Ref. [16]		Ref. [16]	
L $\rightleftarrows$ (Ho) + Ho_2Al	1249	1230	0.240	0.247
L $\rightleftarrows$ Ho_2Al	1291	1295	0.333	0.333
L $\rightleftarrows$ Ho_3Al_2 + Ho_2Al	1257	1281	0.345	0.370
L + HoAl $\rightleftarrows$ Ho_3Al_2	1267	1291	0.360	0.398
L + $HoAl_2$ $\rightleftarrows$ HoAl	1388	1396	0.455	0.467
L $\rightleftarrows$ $HoAl_2$	1803	1802	0.667	0.667
L + $HoAl_2$ $\rightleftarrows$ $HoAl_3$	1360	1366	0.774	0.870
L $\rightleftarrows$ $HoAl_3$ + Al	923	902	0.982	0.967

description of the Gibbs energy functions corresponding to the different models have been presented in a recent paper [7] and are not repeated here. In the following only a few comments are added.

The Gibbs energy of the pure elements refers to the enthalpy of the element in the so-called SER (Standard Element Reference) state (the stable state at 10^5 Pa and 298.15 K).

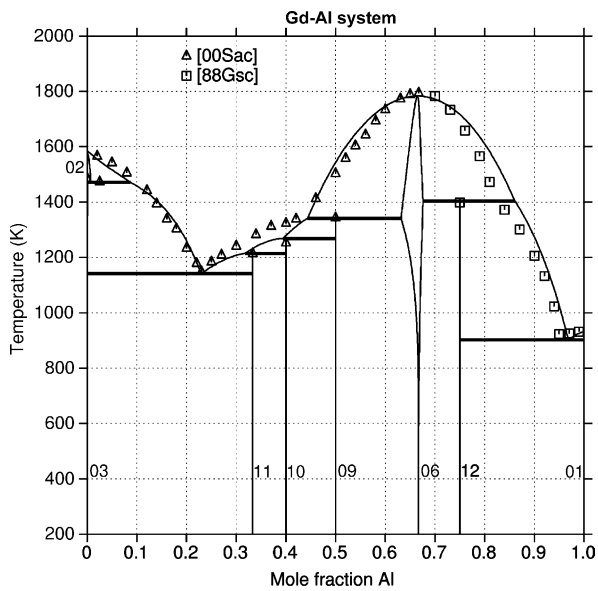

Fig. 2. The computed Gd–Al phase diagram compared with the experimental data points ([00Sac] = [10] and [88Gsc] = [9]). Phases are identified by two-digit labels referring to Tables 1 and 4.

The Gibbs energy functions of the pure elements in the stable phases (A1 and liquid Al, A3, A2 and liquid Gd and Dy, A3 and liquid Ho and Mg) have been taken from the PURE database [34] and are omitted from the Appendix A. The Gibbs energy functions of the pure elements in metastable structures (reference values for the thermodynamic description of the solid solutions) are discussed separately for each phase in the next paragraph.

According to the CEF formalism, the Gibbs energy of the solid phases is expressed as a function of temperature and site fractions by assuming the different atoms to occupy one or more sublattices. The appropriate sublattice model for each solution phase has been selected by taking into account the crystal structure characteristics of the phase itself. In fact, each sublattice is assumed to contain all the atoms having similar interactions with their surroundings. In general, sublattices may correspond to the Wyckoff special positions in the crystal lattice; however, especially in the case of complex structures, in order to reduce the total number of sublattices and, consequently, the number of parameters to be optimised, atoms pertaining to different Wyckoff positions with similar coordination states can be merged in the same sublattice. This problem was discussed, for selected important phases, by Ansara et al. [35] and Ferro and Cacciamani [36] and their conclusions and suggestions are here applied. For each solution phase a concise description of the crystal structure and coordination state is reported in Table 1 together with the sublattice model adopted and a few comments.

In the special case of phases related by an order-disorder transformation, like cI2-W (A2) and cP2-CsCl

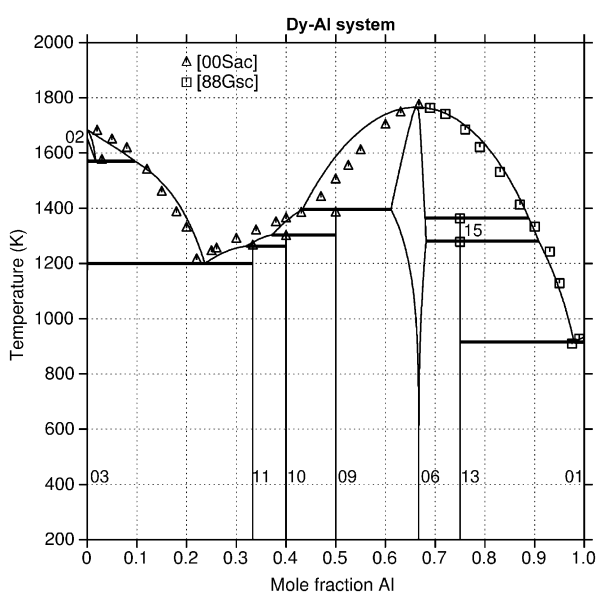

Fig. 3. The computed Dy–Al phase diagram compared with the experimental data points ([00Sac] = [10] and [88Gsc] = [15]). Phases are identified by two-digit labels referring to Tables 1 and 4.

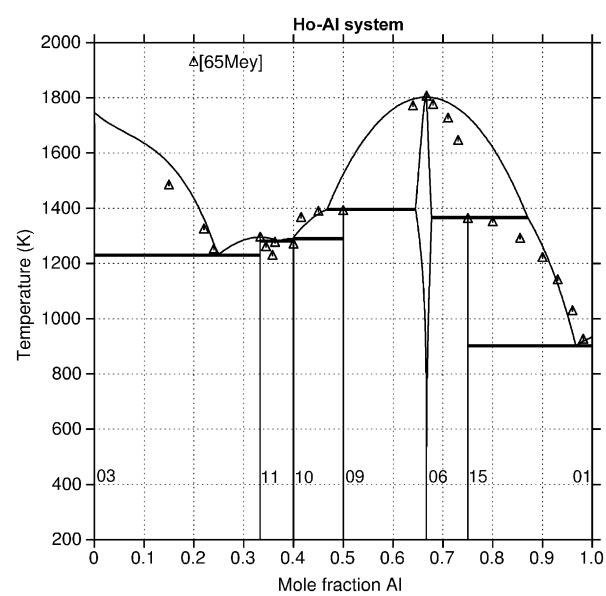

Fig. 4. The computed Ho–Al phase diagram compared to the experimental data points ([65Mey] = [16]). Phases are identified by two-digit labels referring to Tables 1 and 4.

(B2), according to [37], the Gibbs energies of the ordered forms can be related to those of the disordered ones. The resulting model accounts for the ordering contribution to the Gibbs energy and makes the computed phase equilibria more realistic and in better agreement with the experimental data, especially when the phases involved present an extended solid solubility. Owing to the increased complexity of the Gibbs energy expression resulting from this description, the number of optimising parameters is reduced by a number of constraints (mainly related to the phase stoichiometry) introduced and discussed by [37] for the cF4-Cu (A1) and cP4-$AuCu_3$ ($L1_2$) phases and by [38] for cI2-W (A2) and cP2-CsCl (B2).

Finally, it may be useful to highlight that the crystal structures of the γ-MgAl and Dy_5Mg_{24} or Ho_5Mg_{24} binary phases correspond to two different occupation states of the cI58-αMn type structure, often denoted in literature as the A12 or χ phase. This is one of the phases especially discussed by [35] and [36]: their conclusions, which are in complete agreement, have been adopted here. Then the mentioned phases have been modelled as different sublattice occupations of the same cI58-αMn phase (see Table 1).

4. Optimisation

Thermodynamic optimisations and calculations were performed by using the Thermo-Calc software package [39]. For the Al–Mg system the thermodynamic optimisation by [40] was adopted. The phase diagram is here reported in Fig. 1.

The six remaining binary systems have been optimised on the basis of the available experimental data and

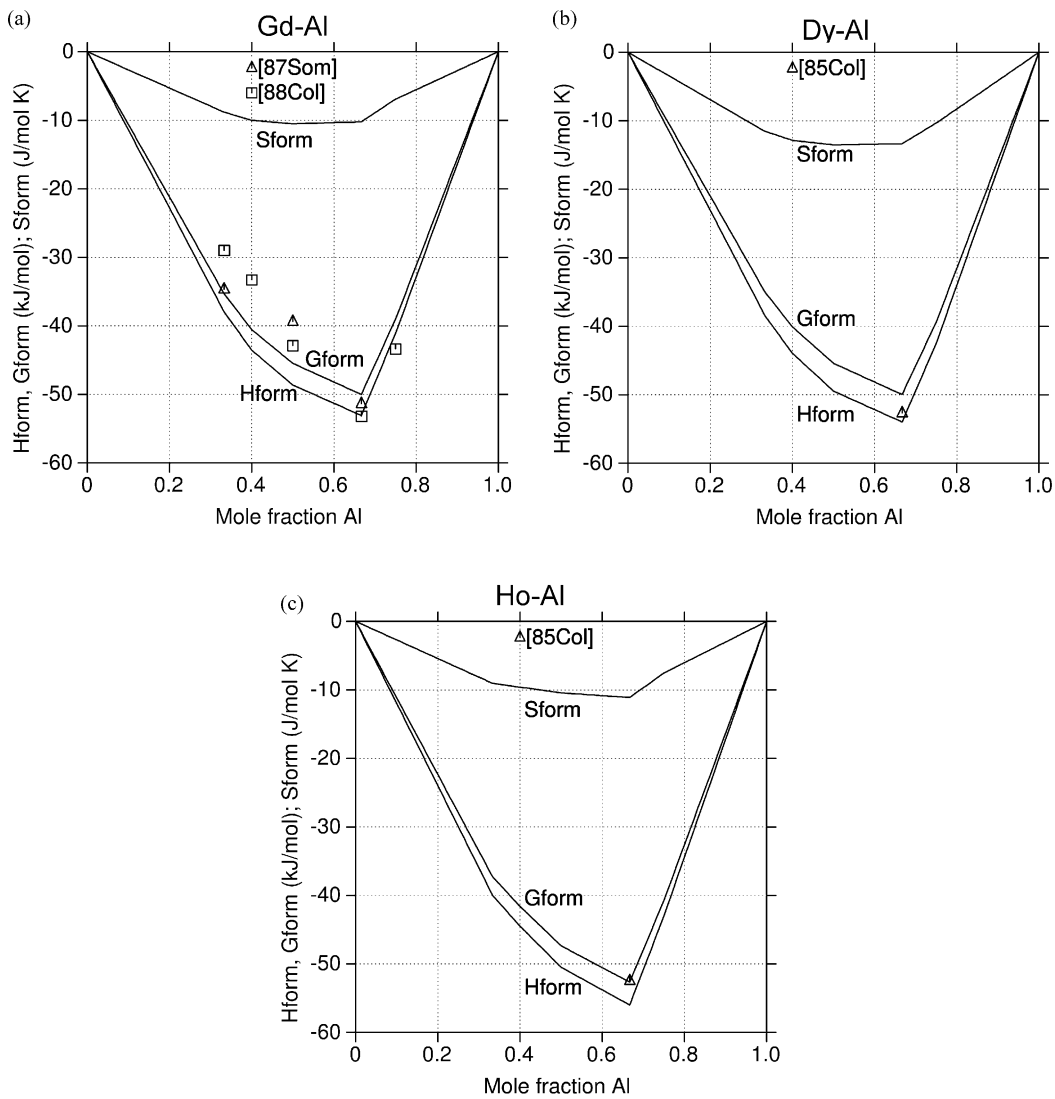

Fig. 5. Computed enthalpies, entropies and Gibbs energies of formation of the solid alloys at 300 K for the Gd–Al, Dy–Al and Ho–Al systems. Experimental enthalpy values are included for comparison ([85Col] = [8], [87Som] = [12], [88Col] = [11]).

considering the close similarity between the rare earths involved. It is well known, in fact, that most properties of the Ln-Metal alloys show smooth and regular variations as a function of the Ln atomic number (especially when trivalent lanthanides are considered): as a consequence the missing values for a specific Ln-Metal alloy may be reliably estimated by interpolation, when the data for other Ln-Metal alloys are available. Interpolated (or extrapolated) parameters from neighbouring systems have then been used as starting values and have been subsequently optimised on the basis of the available experimental data. Once satisfactory optimisations of the binary systems were obtained, tentative calculations of the three ternary systems were performed and recursively refined by comparison with the experimental results. In a few cases it was necessary to re-optimise selected binary parameters in order to be able to reproduce experimentally determined ternary equilibria.

The final thermodynamic description of the six binary and three ternary systems is reported in the Appendix, where parameters optimised in this work are underlined.

In the following a few comments concerning modelling and optimisation of each phase are presented.

4.1. Liquid

As there are no literature data on the binary Al–R and R–Mg liquid phases, parameters in their Gibbs energy functions have been optimized on the basis of the experimental phase equilibria. The start value of these thermodynamic parameters have been estimated by comparison with other Al–R and R–Mg systems, for which some experimental data are available.

Ternary excess parameters were very roughly evaluated (see comments in the following paragraph).

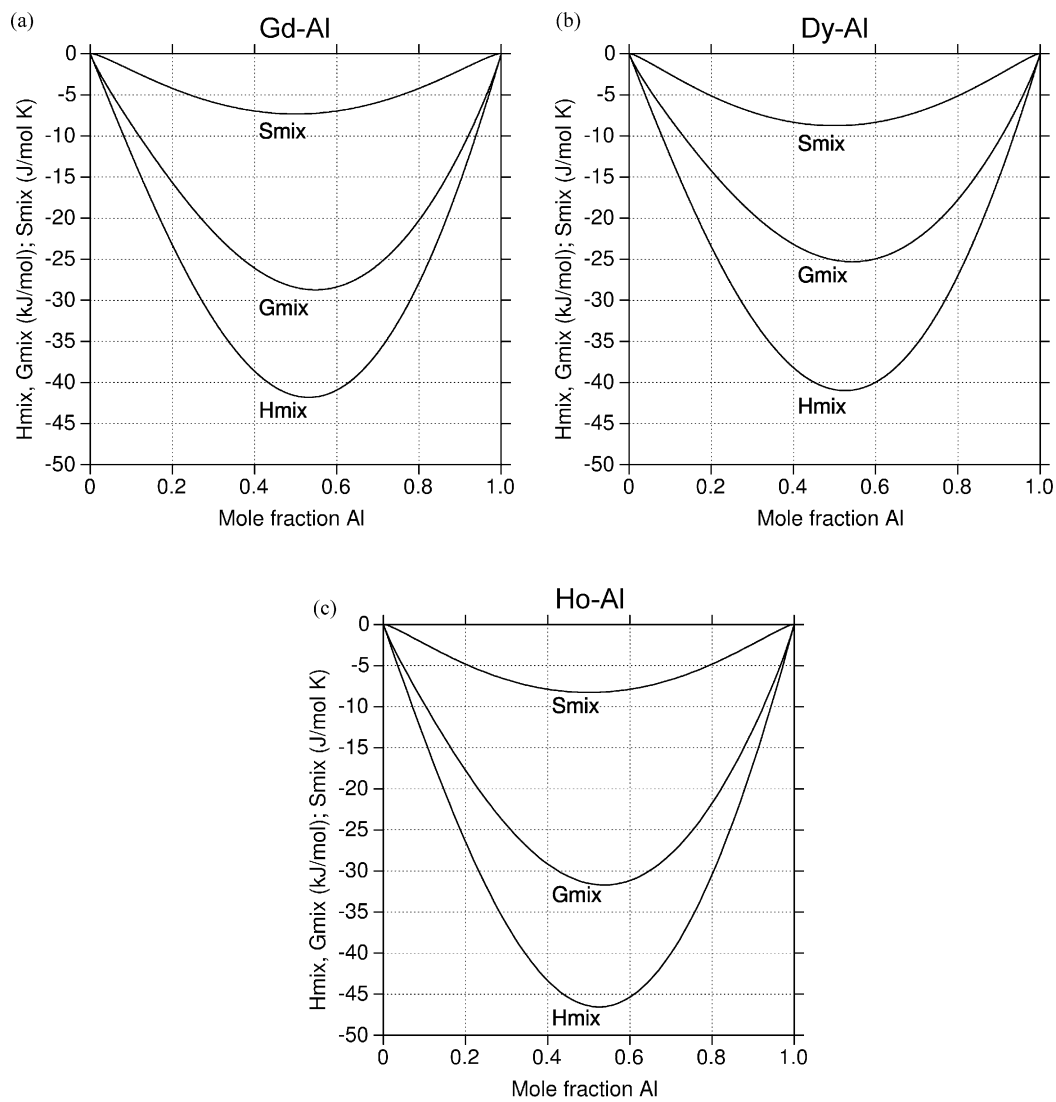

Fig. 6. Thermodynamic properties of mixing of the liquid phase computed at 1800 K for the R–Al systems.

4.2. cF4-Cu (A1)

This is the reference state structure for Al. No data being available in the PURE database [34] for the rare earths considered here, it was assumed:

- $G^{A1}_{Gd} = 1800\text{-}0.1\ R\ T + G^{SER}_{Gd}$
- $G^{A1}_{Dy} = 1300\text{-}0.08\ R\ T + G^{SER}_{Dy}$
- $G^{A1}_{Ho} = 2000\text{-}0.1\ R\ T + G^{SER}_{Ho}$

These were estimated by assuming $\Delta S^{A1-A3} \approx 0.1\ R$, in agreement with Fernandez Guillermet et al. [41], and by evaluating ΔH^{A1-A3} by extrapolating to the pure rare earth the A1 + A3 two-phase field in the binary *R*–Th systems. For Dy the value 0.1 *R* was lowered to 0.08 *R* in order to avoid A2 becoming more stable than A1 at low temperature. No binary R–Mg excess parameters were optimized.

4.3. cI2-W (A2)

The Gibbs energy data for Gd, Dy (stable in this structure at high temperature) and for Al, Mg (not stable in this structure) are reported in the PURE

Table 3
Comparison between experimental and computed invariant equilibria in the optimised R–Al systems

Reaction	T(K) experimental	T(K) calculated	x(Al, liquid) experimental	x(Al, liquid) calculated
Gd–Mg system	Ref. [22]		Ref. [22]	
(β-Gd) ⇄ (α-Gd) + GdMg	973	976	0.260	0.260
L ⇄ (β-Gd) + GdMg	1141	1130	0.505	0.501
L + GdMg ⇄ $GdMg_2$	1029	1029	0.740	0.721
L + $GdMg_2$ ⇄ $GdMg_3$	993	962	0.805	0.799
L + $GdMg_3$ ⇄ $GdMg_5$	931	867	0.840	0.884
L ⇄ $GdMg_5$ + (Mg)	821	836	0.912	0.914
Dy–Mg system	Ref. [27]		Ref. [27]	
(β-Dy) ⇄ (α-Dy) + DyMg	963	958	0.300	0.218
L ⇄ (β-Dy) + DyMg	1133	1130	0.580	0.557
L + DyMg ⇄ $DyMg_2$	983	976	0.780	0.767
L + $DyMg_2$ ⇄ Dy_5Mg_{24}	873	867	0.860	0.865
$DyMg_2$ + Dy_5Mg_{24} ⇄ $DyMg_3$	793	801	0.667	0.681
L ⇄ Dy_5Mg_{24} + (Mg)	833	843	0.905	0.914
Ho–Mg system	Ref. [29]		Ref. [29]	
L + (Ho) ⇄ β-(Ho,Mg)	1443	1465	0.300	0.302
β-(Ho,Mg) ⇄ (Ho) + HoMg	958	958	0.290	0.270
L + β-(Ho,Mg) ⇄ HoMg	1118	1110	0.590	0.593
L + HoMg ⇄ $HoMg_2$	968	967	0.760	0.766
L + $HoMg_2$ ⇄ Ho_5Mg_{24}	873	862	0.860	0.861
L ⇄ Ho_5Mg_{24} + (Mg)	828	842	0.900	0.905

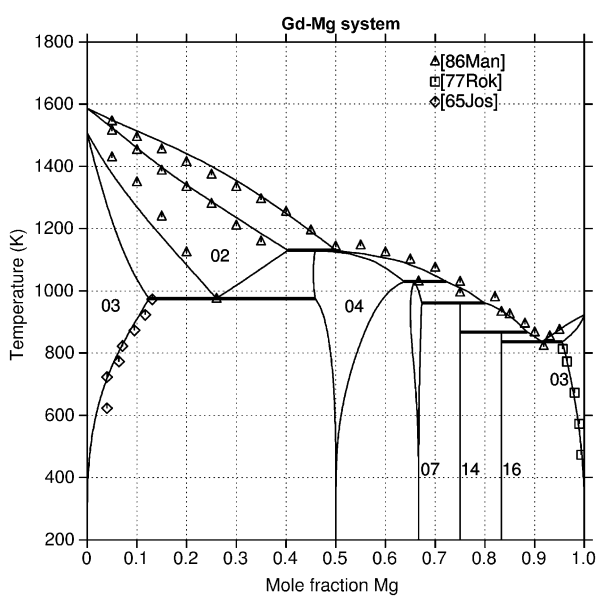

Fig. 7. The computed Gd–Mg phase diagram compared with the experimental data points ([86Man] = [22], [77Rok] = [23] and [65Jos] = [20]). Phases are identified by two-digit labels referring to Tables 1 and 4.

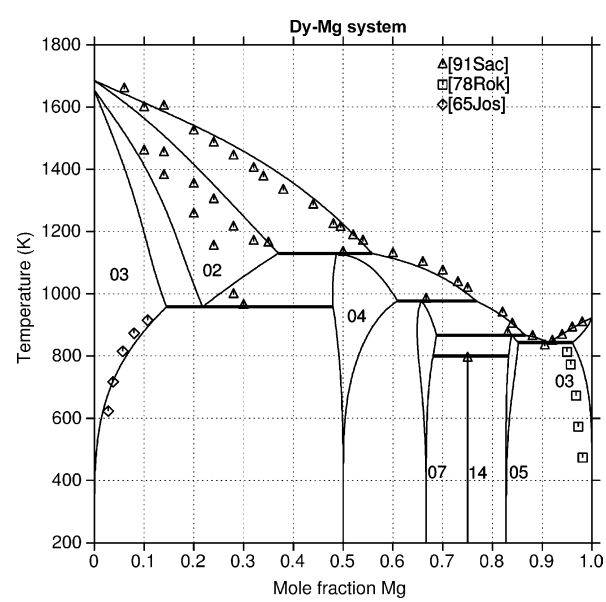

Fig. 8. The computed Dy–Mg phase diagram compared with the experimental data points ([91Sac] = [27], [78Rok] = [28] and [65Jos] = [20]). Phases are identified by two-digit labels referring to Tables 1 and 4.

database [34]. The value of G^{A2}_{Ho} was taken from Norgren [42]. This phase shows a large solubility range in the binary R–Mg systems and binary excess parameters have been optimized also considering the ordering equilibria with the B2 phase.

4.4. hP2-Mg (A3)

This is the reference structure for Gd, Dy, Ho and Mg. The Gibbs energy data for Al in this structure are reported in the PURE database [34]. No binary Al–R excess parameters have been added, because of the negligible Al solubility in hexagonal rare earths. No ternary excess parameters were needed.

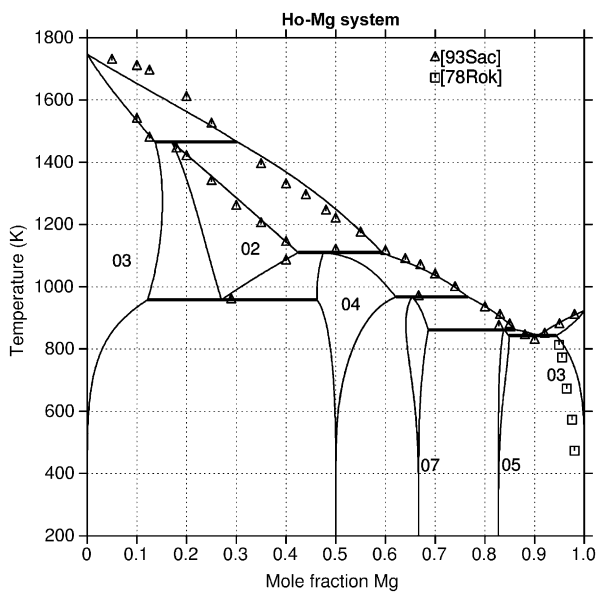

Fig. 9. The computed Ho–Mg phase diagram compared with the experimental data points ([93Sac] = [29], [78Rok] = [30]). Phases are identified by two-digit labels referring to Tables 1 and 4.

4.5. cP2-CsCl (B2)

This phase has been modelled as the ordered form of A2, and the expressions of the binary R–Mg excess functions have been evaluated taking into account the constraints introduced by [38]. In order to reproduce the large ternary solubility of this phase, binary Al–R and ternary excess parameters have been introduced and optimized.

4.6. cI58-αMn (A12)

A12 is present in the Al–Mg system as $Mg_{17}Al_{12}$ and in Dy–Mg and Ho–Mg as R_5Mg_{24}. It does not show any appreciable ternary solubility. According to the model and the site occupations adopted here, it is defined in the x(R) = 0-5/29 and x(Al) = 0-24/29 composition range, not including pure R and Al. The Gibbs energy of pure Mg in the A12 structure is given in the PURE database. The other G terms, not already optimised by [40] and corresponding to the Gibbs energy of formation of hypothetical metastable binary or ternary compounds, were estimated by linear combination of those already determined. The estimation was based on the assumption that, in each sublattice, the enthalpy and entropy of substitution of one atom by another are only dependent on the nature of the two atoms and independent on composition, temperature and occupation of the other sublattices.

4.7. Laves phases: cF24-$MgCu_2$ (C15), hP12-$MgZn_2$ (C14), hP24-$MgNi_2$ (C36)

These three phases have been modelled according to the assumptions generally adopted for the Laves phases (see [35] and [36]). The cF24-$MgCu_2$ phase is stable in all the Al–R systems as RAl_2 and dissolves appreciable amounts of Mg by substitution of Al: this required the introduction of metastable binary R–Mg parameters for

Table 4
Summary of the sublattice models adopted for the phases considered as stoichiometric in the systems optimised. Phase labels are used for phase identification in the figures

Phase labels	Phase identification according to the structure	Phase identification according to the composition	Sublattice model (R = Gd,Dy,Ho)
09	oP16-DyAl	GdAl, DyAl, HoAl	$(R)_1(Al)_1$
10	tP20-Zr_3Al_2	Gd_3Al_2, Dy_3Al_2, Ho_3Al_2	$(R)_3(Al)_2$
11	oP12-Co_2Si	Gd_2Al, Dy_2Al, Ho_2Al	$(R)_2(Al)_1$
12	hP8-antiNi_3Sn	$GdAl_3$	$(Al)_3(Gd)_1$
13	hP16-$TiNi_3/DO_{24}$	$DyAl_3$(rt)	$(Dy)_1(Al)_3$
14	cF16-BiF_3	$GdMg_3$, $DyMg_3$	$(R)_1(Mg)_3$
15	hR60-$HoAl_3$	$DyAl_3$(ht), $HoAl_3$	$(R)_1(Al)_3$ (R = Dy,Ho)
16	cF448-$GdMg_5$	$GdMg_5$	$(Gd)_1(Mg)_5$
17	cF1832-Mg_2Al_3	β-(Mg,Al) or Mg_2Al_3	$(Mg)_{89}(Al)_{140}$
18	hR159-$Co_5Cr_2Mo_3$	ϵ-(Mg,Al)	$(Mg)_{23}(Al)_{30}$

Dy and Ho. This structure is also stable in the Gd–Mg system as $GdMg_2$: the introduction of a ternary interaction parameter was needed in order to obtain the correct solubility limit on the Al–Gd side of the Al–Gd–Mg ternary system.

The hP12-$MgZn_2$ phase is stable at the RMg_2 composition in the remaining R-Mg systems; its very small Al solubility was not considered.

Finally, the hP24-$MgNi_2$ phase is stable in the ternary systems as a ternary phase at the approximate composition $Al_2(R_xMg_{(1-x)})$ with x = 0.36 to 0.5, depending on R: it shows a small solubility range probably due to the substitution between R and Mg. Binary metastable parameters were estimated by analogy with other Laves phases and ternary parameters were optimised in order to obtain the desired stability ranges in temperature and composition.

4.8. Other phases

All the remaining phases β-(Mg,Al), ϵ-(Mg,Al), R_2Al-Co_2Si, R_3Al_2-Zr_3Al_2, RAl-DyAl, RAl_3-$HoAl_3$ (R = Dy, Ho), RAl_3-$TiNi_3$ (R = Dy), Al_3R-Ni_3Sn (R = Gd), RMg_3-BiF_3 (R = Gd, Dy) and $GdMg_5$ have been described as binary stoichiometric compounds.

5. Results and discussion

The Gibbs energy functions of all the phases in the optimised binary and ternary systems are summarised in the Appendix, where the underlined parameters are those evaluated in this work. The derived phase equilibria and thermodynamic functions are presented and briefly discussed in the following with reference not only

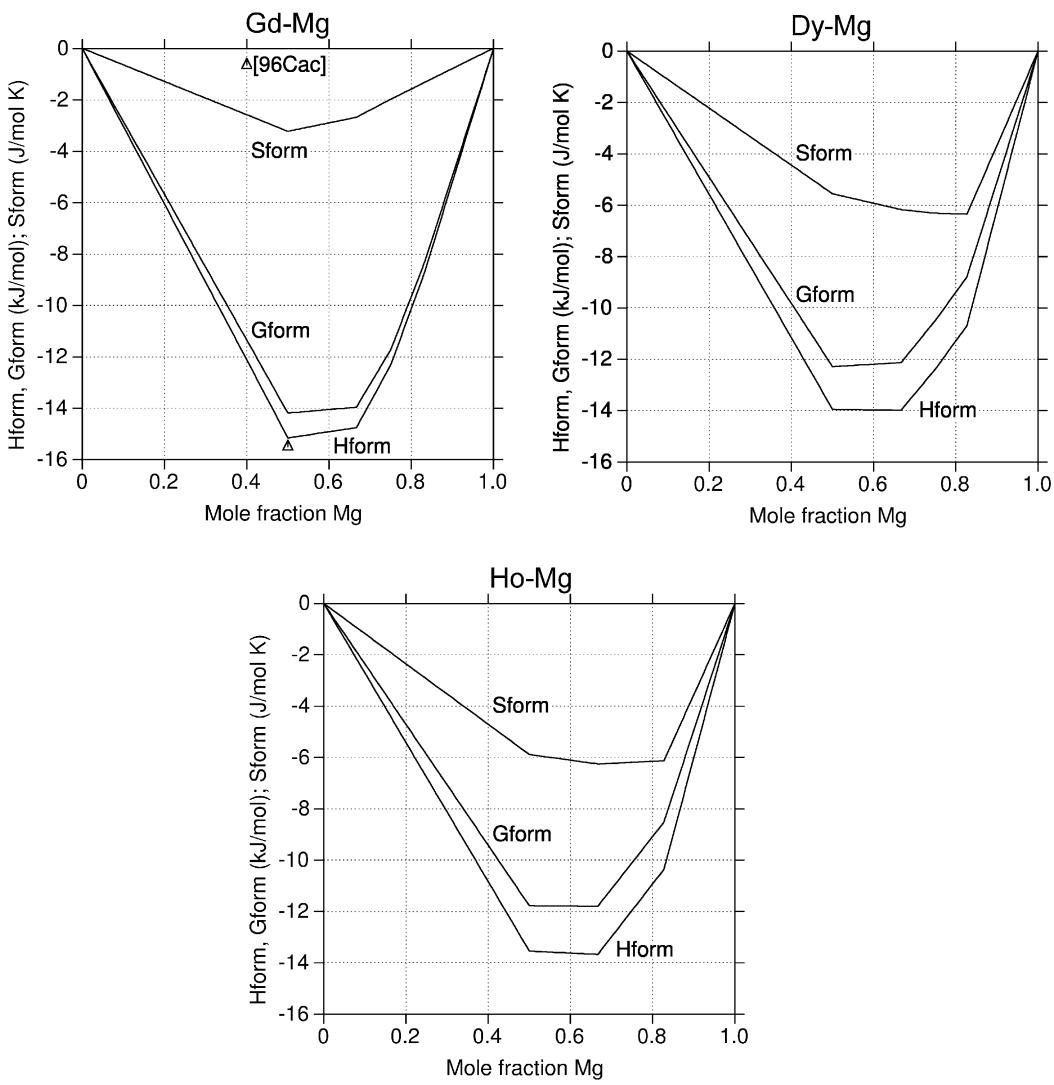

Fig. 10. Computed enthalpies, entropies and Gibbs energies of formation of the solid alloys at 300 K for the Gd–Mg, Dy–Mg and Ho–Mg systems. Experimental enthalpy values are included for comparison ([96Cac] = [25]).

to the available literature data but also to the smooth and regular variations expected when considering systems involving different rare earths.

5.1. Al–Gd, Al–Dy and Al–Ho systems

The optimised Al–Gd, Al–Dy and Al–Ho phase diagrams are shown in Figs. 2–4 respectively. Experimental and computed invariant equilibria are also compared in Table 2.

The agreement between experimental data and computed equilibria is better for Al-Gd and Al-Dy, which were experimentally re-investigated more recently [10], than for Al-Ho, not reinvestigated after [16]. The calculated Al-Ho, however, shows good agreement with homologous Gd and Dy systems. In the experimental Al-Ho phase diagram quite questionable liquidus curvatures may be observed. On the basis of these considerations and of the results obtained in the Al–Ho–Mg ternary system, the computed Al–Ho system has been considered satisfactorily modelled. In the three computed systems the Laves phase shows an appreciable solubility range at high temperature: it is a reasonable behaviour but there is no experimental indication of that. It would be necessary to change the phase model or to add more excess parameters in order to reduce such a solubility range: this was not considered opportune and the solubility ranges were kept.

Experimental and computed enthalpies of formation of the solid alloys at room temperature are compared in Fig. 5 where computed Gibbs energies and entropies are also reported. Thermodynamic properties of mixing of the liquid phases computed at 1800 K are reported in Fig. 6. The partial mixing enthalpy of Gd in liquid Al at infinite dilution measured by [13] is in good agreement with the optimised thermodynamics;

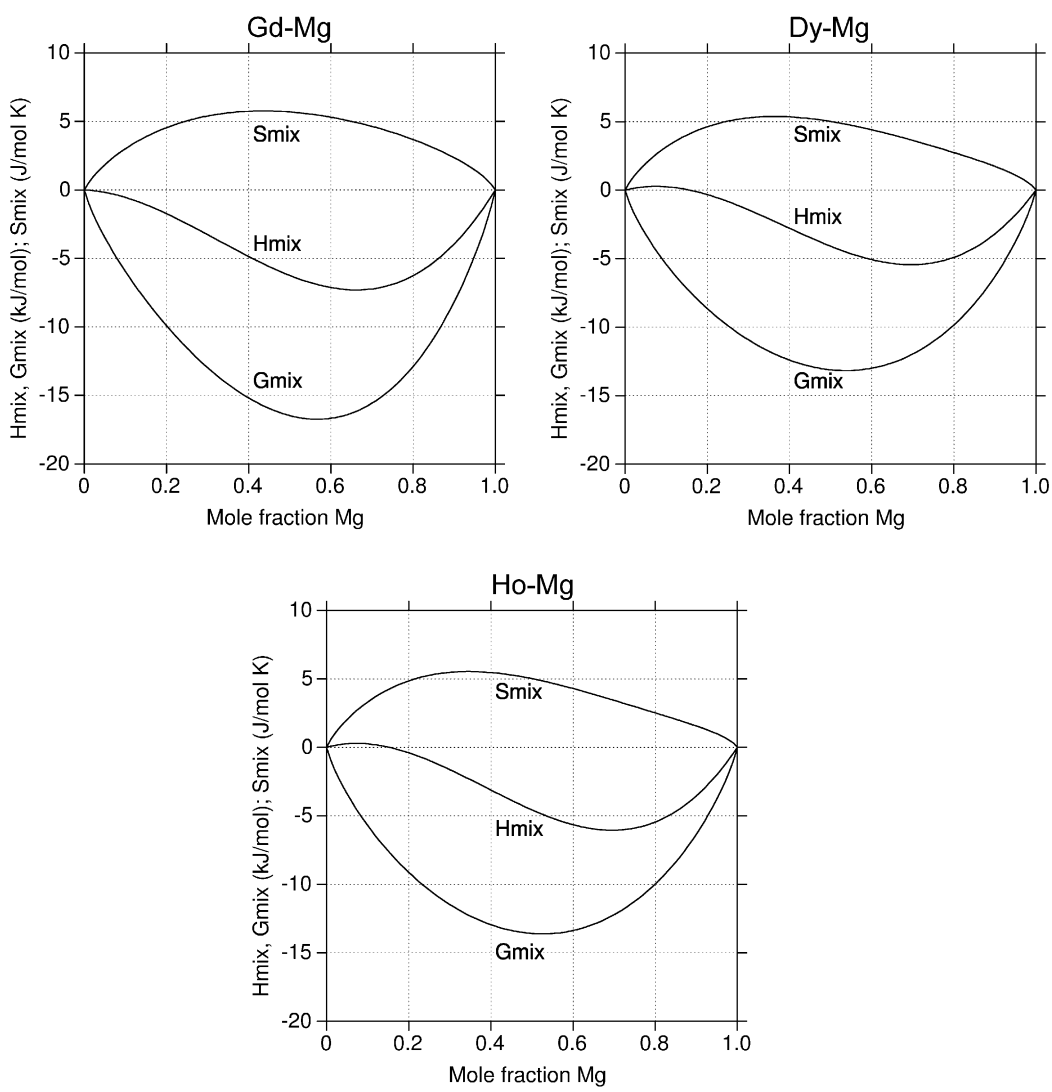

Fig. 11. Thermodynamic properties of mixing of the liquid phase computed at 1800 K for the R–Mg systems.

however the measurements of the Gd chemical potential carried out by [14] seem inconsistent with other experimental results and have been neglected in the last optimisation steps.

Qualitative agreement with the Al–Gd–Mg phase diagram computed by [5] may be observed. A more significant comparison is difficult because [5] did not report the phase models and the values of the optimised parameters. Anyway he did not consider the more recent phase diagram investigation carried out by [10].

5.2. Gd–Mg, Dy–Mg and Ho–Mg systems

Optimised Gd-Mg, Dy–Mg and Ho–Mg phase diagrams are reported and compared with the experimental data in Figs. 7–9. Experimental and computed invariant equilibria are also compared in Table 3.

General good agreement and a few discrepancies may be noticed between computed and experimentally observed phase equilibria.

In the Gd–Mg system phase modelling and optimisation have been improved with respect to our previous work [26]. Nevertheless small differences in the peritectic decomposition temperatures of $GdMg_3$ and $GdMg_5$ (see Table 3) remain. However, the shape of the experimental phase diagram is rather improbable in this region.

In the Dy–Mg system the composition of the A2 phase at the $A2 \leftrightarrows B2 + A3$ eutectoidic equilibrium shows an appreciable difference and in both Dy–Mg and Ho–Mg the rare earth solid solubility in Mg seems to decrease with temperature more rapidly than according to [28] and [30] respectively.

In all three systems an appreciable, but not defined, solid solubility range is reported for the B2 phase. In the optimisation of this phase the smallest number of parameters required to reproduce the invariant equilibria and the liquidus curve was used and the consequent solid solubility accepted.

Thermodynamic functions of formation at room temperature are reported in Fig. 10. A good agreement may be noticed between computed and experimental enthalpy of formation by [25], which is also consistent with [18]. The results by [19], however, seem less reliable and have not been considered in the final optimisations. Mixing functions of the liquid phase, which cannot be compared with any experimental data, are reported in Fig. 11.

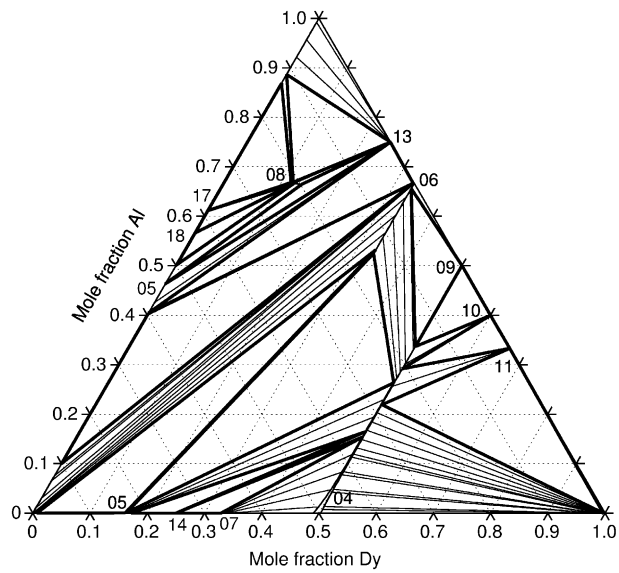

Fig. 13. Al–Dy–Mg system: isothermal section computed at 673 K (400 °C). Phases are identified by two-digit labels referring to Tables 1 and 4.

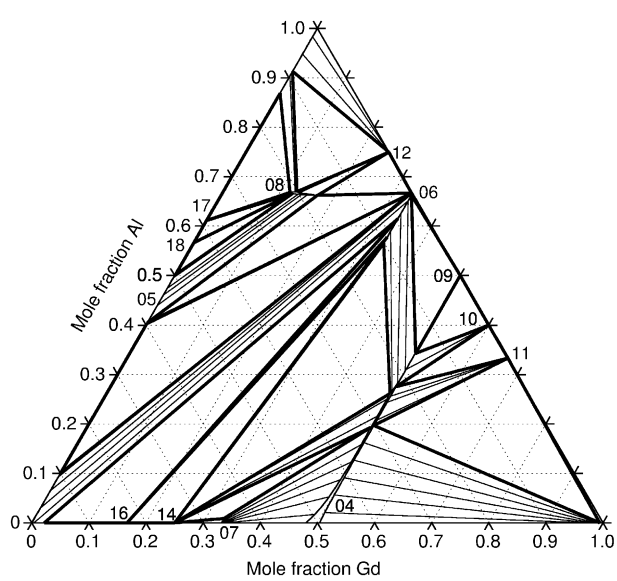

Fig. 12. Al–Gd–Mg system: isothermal section computed at 673 K (400 °C). Phases are identified by two-digit labels referring to Tables 1 and 4.

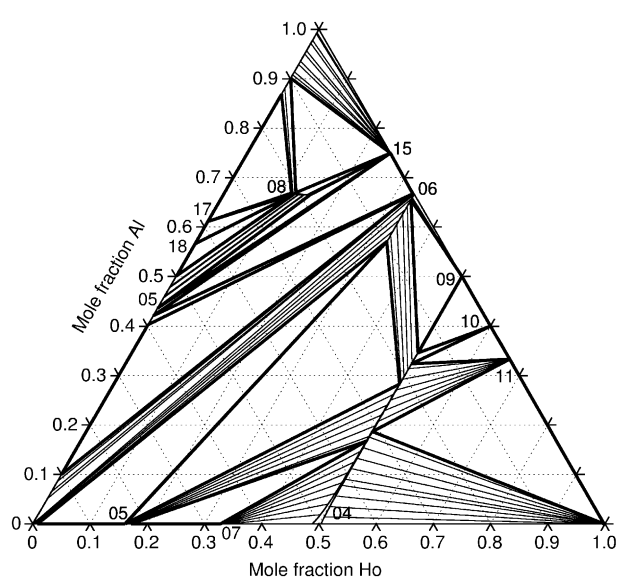

Fig. 14. Al–Ho–Mg system: isothermal section computed at 673 K (400 °C). Phases are identified by two-digit labels referring to Tables 1 and 4.

Ab-initio calculations by [21] are in rough agreement with present results only for the enthalpies of mixing of liquid and disordered solid Gd–Mg alloys.

5.3. Al–Gd–Mg, Al–Dy–Mg and Al–Ho–Mg systems

The optimised Al–Gd–Mg, Al–Dy–Mg and Al–Ho–Mg isothermal sections at 673 K (400 °C) are shown in Figs. 12–14. They are in excellent agreement with the experimental results reported in Figs. 1,3 and 6 of part I [1], respectively. Al-Gd-Mg, is in good agreement also with the investigation by [31] except for the decomposition temperature of the C36 ternary phase: on the basis of our investigations (see part I) and considering the behaviour observed in the other systems a decomposition temperature around 450 °C was preferred to the higher value suggested by [31]. A higher stability of C36, indeed, seems inconsistent with the phase equilibria determined at 400 °C.

A second series of isothermal sections, calculated at 1073 K (800 °C), are shown in Fig. 15. The solid solubilities of the C15 phase, appreciably larger than those at 673 K, are confirmed, for Al–Gd–Mg, by both [43] and part I.

For the reasons explained in the previous paragraph a full comparison with the Al–Gd–Mg optimisation performed by [5] is not possible. According to the isothermal section shown in that paper no ternary solid solubility was taken into account: as a consequence only partial agreement with the present calculation can be observed (see part I for a more detailed comparison between [5] and our experimental results).

The projections of the liquidus surfaces are reported in Figs. 16–18 (notice that invariant equilibria involving the Al–Mg binary phases are not visible in the figure because they are very close to the Al–Mg binary subsystem). The extents of the primary solidification regions for the different phases are in good agreement with the experimental information reported in part I. As expected, considering its high stability, a large area of the ternary field is dominated by the C15 phase.

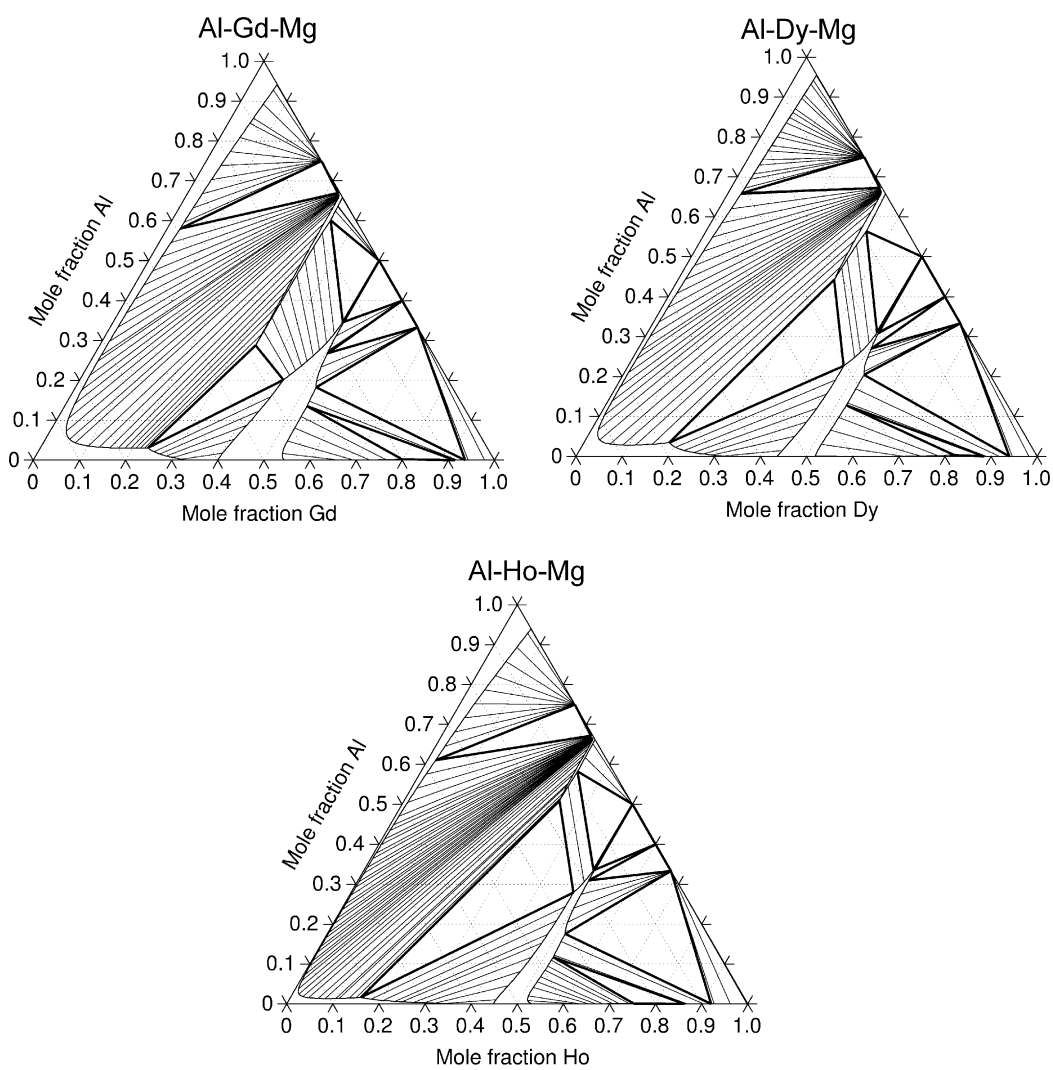

Fig. 15. Al–Gd–Mg, Al–Dy–Mg and Al–Ho–Mg isothermal section computed at 1073 K (800 °C).

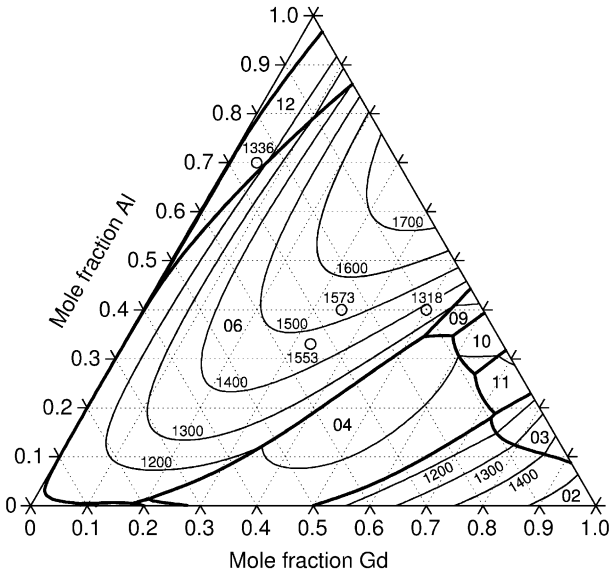

Fig. 16. Al–Gd–Mg liquidus surface. DTA results obtained in part I [1] are included for comparison. Selected primary phases are identified by two-digit labels referring to Tables 1 and 4.

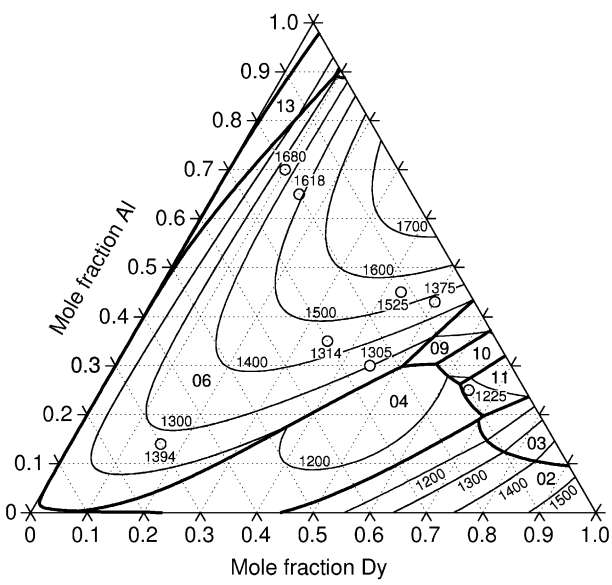

Fig. 17. Al–Dy–Mg liquidus surface. DTA results obtained in part I [1] are included for comparison. Selected primary phases are identified by two-digit labels referring to Tables 1 and 4.

The liquidus temperatures experimentally determined by DTA are also reported in the Figs. 16–18. A qualitative agreement may be observed but some experimental results, especially those from samples richer in Al (and with higher liquidus temperatures), seem to be in poor agreement with the computed ternary liquidus and also with the close binary Al–R systems. Such discrepancies can be caused by side reactions due to the high temperature reactivity of the Al-rich alloys. Considering the limited reliability of the experiments, the ternary interaction parameters of the liquid phase were only roughly evaluated.

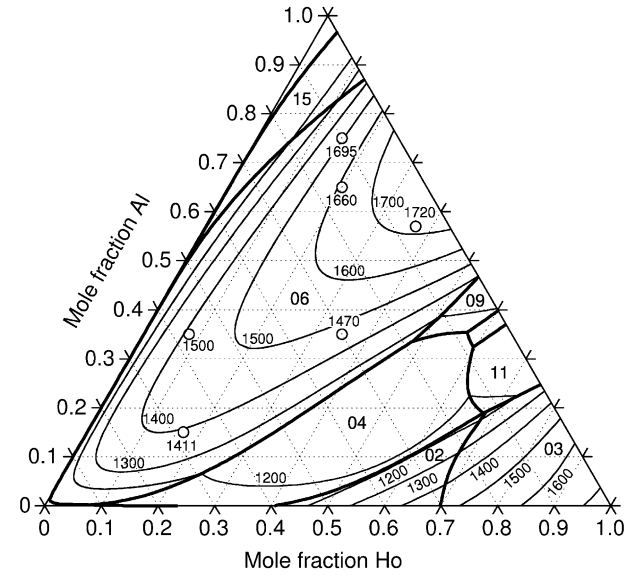

Fig. 18. Al–Ho–Mg liquidus surface. DTA results obtained in part I [1] are included for comparison. Selected primary phases are identified by two-digit labels referring to Tables 1 and 4.

Some uncertainty affects the ternary equilibria involving the A2 phase at high temperature and in the R-rich part of the diagrams: the close structural relation with B2, which dissolves a large amount of Al, would suggest a correspondingly appreciable Al solubility in A2. In the present calculation, however, as no experimental evidence is available, the Al solubility in the A2 phase was not considered.

6. Conclusion

Three Al–R and three R–Mg (R = Gd, Dy, Ho) binary systems have been optimised by using the experimental data available in literature. Then, on the basis of the experimental results obtained in part I [1] the three Al–R–Mg phase diagrams were optimised. The reliability of the optimisation results is supported by the general good agreement between computed and experimental data concerning both phase equilibria and thermodynamics. Some uncertainty concerns temperatures of the solid-liquid ternary equilibria, due to uncertainties on the experimental DTA results, and the ternary solubility of the R–Mg based A2 phase at temperatures higher than that of the investigated isothermal section. For most phases the optimisation required no or small ternary interaction parameters: this supports the assumption that, in many cases, a ternary phase diagram calculated by a combination of the binary ones (when these are well established and modelled) may be a reasonably good approximation of the real ternary phase equilibria, especially when thermodynamics of the metastable end-members of the solution phases can be estimated by analogy with similar stable phases.

The application of the close analogies and regularities relating the characteristics of the rare earth alloy systems and the use of a recursive approach (experiments and thermodynamic modelling) allowed us to obtain reliable thermodynamic descriptions of quite complex ternary systems on the basis of a relatively small number of samples experimentally investigated.

Acknowledgements

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Appendix A

Parameters evaluated in this work are underlined. Binary Al–Mg functions and parameters are taken from [40].

A.1. Functions

GHSERAL, GHSERGD, GHSERDY, GHSERHO, GHSERMG, GLIQAL, GLIQGD, GLIQDY, GLIQHO, GLIQMG, GBCCGD, GBCCDY are taken from [34]

GLAVAL = $\underline{+15000 + 3\ \text{GHSERAL}}$
GLAVGD = $\underline{+15000 + 3\ \text{GHSERGD}}$
GLAVDY = $\underline{+15000 + 3\ \text{GHSERDY}}$
GLAVHO = $\underline{+15000 + 3\ \text{GHSERHO}}$
GLAVMG = $\underline{+15000 + 3\ \text{GHSERMG}}$
GA12MG = $+133469 - 87.319\ T$
GA12ALMG = $-52780 - 50.75\ T$
GA12MGAL = $+97875 - 101.5\ T$
GA12GDMG = $\underline{-50000 + 50\ T}$
GA12DYMG = $\underline{-310000 + 184.6\ T}$
GA12HOMG = $\underline{-300000 + 177.7\ T}$
GC14GDMG = $\underline{\text{GC15GDMG} + 15000}$
GC14DYMG = $\underline{-42000 + 18.66\ T}$
GC14HOMG = $\underline{-41000 + 18.75\ T}$
GC15ALGD = $\underline{-160000 + 32.64\ T}$
GC15ALDY = $\underline{-162000 + 40.24\ T}$
GC15ALHO = $\underline{-168000 + 33.34\ T}$
GC15GDMG = $\underline{-45000 + 9.972\ T}$
GC15DYMG = $\underline{\text{GC14DYMG} + 11000 - 1.6\ T}$
GC15HOMG = $\underline{\text{GC14HOMG} + 15000 - 1.6\ T}$
GC36ALGD = $\underline{\text{GC15ALGD} + 13000}$
GC36ALDY = $\underline{\text{GC15ALDY} + 15000}$
GC36ALHO = $\underline{\text{GC15ALHO} + 15000}$
GC36GDMG = $\underline{\text{GC15GDMG} + 15000}$
GC36DYMG = $\underline{\text{GC14DYMG} + 15000}$
GC36HOMG = $\underline{\text{GC14HOMG} + 15000}$

A.2. Solution phases

Liquid: Model (Al,R,Mg)

Unary parameters
$\Delta_f G(\text{Liquid,Al;0}) = +\text{GLIQAL}$
$\Delta_f G(\text{Liquid,Gd;0}) = \text{GLIQGD}$
$\Delta_f G(\text{Liquid,Dy;0}) = \text{GLIQDY}$
$\Delta_f G(\text{Liquid,Ho;0}) = +121.2496 + \text{GLIQHO}$
$\Delta_f G(\text{Liquid,Mg;0}) = +\text{GLIQMG}$

Binary parameters
L(Liquid,Al,Gd;0) = $\underline{-166500 + 52.36\ T}$
L(Liquid,Al,Gd;1) = $\underline{-23790 + 0.0\ T}$
L(Liquid,Al,Gd;2) = $\underline{+18520}$
L(Liquid,Al,Dy;0) = $\underline{-163500 + 58.07\ T}$
L(Liquid,Al,Dy;1) = $\underline{-18300 + 0.0\ T}$
L(Liquid,Al,Dy;2) = $\underline{+16130}$
L(Liquid,Al,Ho;0) = $\underline{-185720 + 56.18\ T}$
L(Liquid,Al,Ho;1) = $\underline{-20820 + 0.0\ T}$
L(Liquid,Al,Ho;2) = $\underline{+22050}$
L(Liquid,Gd,Mg;0) = $\underline{-25000 + 0.3089\ T}$
L(Liquid,Gd,Mg;1) = $\underline{+23600 - 4.521\ T}$
L(Liquid,Dy,Mg;0) = $\underline{-16380 + 3.006\ T}$
L(Liquid,Dy,Mg;1) = $\underline{+23860 - 9.803\ T}$
L(Liquid,Ho,Mg;0) = $\underline{-18280 + 3.002\ T}$
L(Liquid,Ho,Mg;1) = $\underline{+26340 - 12.17\ T}$
L(Liquid,Al,Mg;0) = $-12000 + 8.566\ T$
L(Liquid,Al,Mg;1) = $+1894 - 3\ T$
L(Liquid,Al,Mg;2) = $+2000$

Ternary parameters
L(Liquid,Al,Gd,Mg;0) = $\underline{-20000}$
L(Liquid,Al,Dy,Mg;0) = $\underline{-40000}$
L(Liquid,Al,Ho,Mg;0) = $\underline{-20000}$

A1: Model (Al,R,Mg)

Unary parameters
$\Delta_f G(\text{A1,Al;0}) = +\text{GHSERAL}$
$\Delta_f G(\text{A1,Gd;0}) = \underline{+1800 - 0.1\ R\ T + \text{GHSERGD}}$
$\Delta_f G(\text{A1,Dy;0}) = \underline{+1300 - 0.08\ R\ T + \text{GHSERDY}}$
$\Delta_f G(\text{A1,Ho;0}) = \underline{+2000 - 0.1\ R\ T + \text{GHSERHO}}$
$\Delta_f G(\text{A1,Mg;0}) = +2600 - 0.9\ T + \text{GHSERMG}$
L(A1,Al,Mg;0) = $+4971 - 3.5\ T$
L(A1,Al,Mg;1) = $+900 + 0.423\ T$
L(A1,Al,Mg;2) = $+950$

A2: Model (Al,R,Mg)
This is also the disordered part of A2_B2

Unary parameters
$\Delta_f G(\text{A2,Al;0}) = +10083 - 4.813\ T + \text{GHSERAL}$
$\Delta_f G(\text{A2,Gd;0}) = +\text{GBCCGD}$

$\Delta_f G(A2,Dy;0) = +GBCCDY$
$\Delta_f G(A2,Ho;0) = +4380-0.3\ R\ T+GHSERHO$
$\Delta_f G(A2,Mg;0) = +3100-2.1\ T+GHSERMG$

Binary parameters
$L(A2,Al,Gd;0) = \underline{-80000+30.64\ T}$
$L(A2,Al,Dy;0) = \underline{-82000+30.64\ T}$
$L(A2,Al,Ho;0) = \underline{-70000+17\ T}$
$L(A2,Gd,Mg;0) = \underline{-23000}$
$L(A2,Gd,Mg;1) = \underline{+14760}$
$L(A2,Dy,Mg;0) = \underline{-11260+1.700\ T}$
$L(A2,Dy,Mg;1) = \underline{+3755}$
$L(A2,Ho,Mg;0) = \underline{-15960+4.196\ T}$
$L(A2,Ho,Mg;1) = \underline{+4563}$

B2: Model $(Al,R,Mg)_{0.5}(Al,R,Mg)_{0.5}$ (disordered contribution from A2)

Binary parameters
$L(B2,Al:Gd;0) = \underline{-64000+12.0\ T}$
$L(B2,Gd:Al;0) = \underline{-64000+12.0\ T}$
$L(B2,Al:Dy;0) = \underline{-64000+19.0\ T}$
$L(B2,Dy:Al;0) = \underline{-64000+19.0\ T}$
$L(B2,Al:Ho;0) = \underline{-75000+20.00\ T}$
$L(B2,Ho:Al;0) = \underline{-75000+20.00\ T}$
$L(B2,Gd:Mg;0) = \underline{-15380+12.61\ T}$
$L(B2,Mg:Gd;0) = \underline{-15380+12.61\ T}$
$L(B2,Gd,Mg:Gd;0) = \underline{+20960}$
$L(B2,Gd:Gd,Mg;0) = \underline{+20960}$
$L(B2,Dy:Mg;0) = \underline{-15810+13.07\ T}$
$L(B2,Mg:Dy;0) = \underline{-15810+13.07\ T}$
$L(B2,Dy,Mg:Dy;0) = \underline{+22260}$
$L(B2,Dy:Dy,Mg;0) = \underline{+22260}$
$L(B2,Ho:Mg;0) = \underline{-17110+14.26\ T}$
$L(B2,Mg:Ho;0) = \underline{-17110+14.26\ T}$
$L(B2,Ho,Mg:Ho;0) = \underline{+18930}$
$L(B2,Ho:Ho,Mg;0) = \underline{+18930}$

Ternary parameters
$L(B2,Gd:Al,Mg;0) = \underline{-50000+36\ T}$
$L(B2,Al,Mg:Gd;0) = \underline{-50000+36T}$
$L(B2,Dy:Al,Mg;0) = \underline{-52000+34\ T}$
$L(B2,Al,Mg:Dy;0) = \underline{-52000+34\ T}$
$L(B2,Ho:Al,Mg;0) = \underline{-45000+28\ T}$
$L(B2,Al,Mg:Ho;0) = \underline{-45000+28\ T}$

A3: Model (Al,R,Mg)

Unary parameters
$\Delta_f G(A3,Al;0) = +5481-1.8\ T+GHSERAL$
$\Delta_f G(A3,Gd;0) = +GHSERGD$
$\Delta_f G(A3,Dy;0) = +GHSERDY$
$\Delta_f G(A3,Ho;0) = +GHSERHO$
$\Delta_f G(A3,Mg;0) = +GHSERMG$

Binary parameters
$L(A3,Gd,Mg;0) = \underline{-20730+7.882\ T}$
$L(A3,Gd,Mg;1) = \underline{+10660}$
$L(A3,Dy,Mg;0) = \underline{-6302+2.5\ T}$
$L(A3,Dy,Mg;1) = \underline{+5151}$
$L(A3,Ho,Mg;0) = \underline{+5320-9.228\ T}$
$L(A3,Ho,Mg;1) = \underline{+9079}$
$L(A3,Al,Mg;0) = +1950-2\ T$
$L(A3,Al,Mg;1) = +1480-2.08\ T$
$L(A3,Al,Mg;2) = +3500$

A12: Model $(R,Mg)_5(Al,Mg)_{12}(Al,Mg)_{12}$

Unary parameters
$\Delta_f G(A12,Mg:Mg:Mg;0) = +GA12MG$
$+29\ GHSERMG$

Binary parameters
$\Delta_f G(A12,Gd:Al:Al;0) = \underline{+GA12GDMG}$
$\underline{+GA12MGAL-GA12MG}$
$\Delta_f G(A12,Dy:Al:Al;0) = \underline{+GA12DYMG+}$
$\underline{+GA12MGAL-GA12MG}$
$\Delta_f G(A12,Ho:Al:Al;0) = \underline{+GA12HOMG+}$
$\underline{+GA12MGAL-GA12MG}$
$\Delta_f G(A12,Gd:Mg:Mg;0) = GA12GDMG$
$\Delta_f G(A12,Dy:Mg:Mg;0) = GA12DYMG$
$\Delta_f G(A12,Ho:Mg:Mg;0) = GA12HOMG$
$\Delta_f G(A12,Mg:Mg:Al;0) = +GA12ALMG$
$\Delta_f G(A12,Mg:Al:Al;0) = +GA12MGAL$
$\Delta_f G(A12,Mg:Al:Mg;0) = +GA12MGAL+$
$+GA12MG-GA12ALMG$
$L(A12,Mg:Al:Al,Mg;0) = +113100-14.5\ T$

Ternary parameters
$\Delta_f G(A12,Gd:Mg:Al;0) = \underline{+GA12GDMG+}$
$\underline{-GA12MG+GA12ALMG}$
$\Delta_f G(A12,Gd:Al:Mg;0) = \underline{+GA12GDMG+}$
$\underline{+GA12MGAL}$
$\underline{-GA12ALMG}$
$\Delta_f G(A12,Dy:Mg:Al;0) = \underline{+GA12DYMG+}$
$\underline{-GA12MG+GA12ALMG}$
$\Delta_f G(A12,Dy:Al:Mg;0) = \underline{+GA12DYMG+}$
$\underline{+GA12MGAL}$
$\underline{-GA12ALMG}$
$\Delta_f G(A12,Ho:Mg:Al;0) = \underline{+GA12HOMG+}$
$\underline{-GA12MG+GA12ALMG}$
$\Delta_f G(A12,Ho:Al:Mg;0) = \underline{+GA12HOMG}$
$\underline{+GA12MGAL}$
$\underline{-GA12ALMG}$

C14: Model $(Al,R,Mg)_1(Al,R,Mg)_2$

Unary parameters
$\Delta_f G(C14,Al:Al;0) = +GLAVAL$

$\Delta_f G(C14,Gd:Gd;0) = +GLAVGD$
$\Delta_f G(C14,Dy:Dy;0) = +GLAVDY$
$\Delta_f G(C14,Ho:Ho;0) = +GLAVHO$
$\Delta_f G(C14,Mg:Mg;0) = +GLAVMG$

Binary parameters
$\Delta_f G(C14,Gd:Mg;0) = \underline{+GC14GDMG}$
$\Delta_f G(C14,Mg:Gd;0) = \underline{-GC14GDMG + 30000}$
$\Delta_f G(C14,Dy:Mg;0) = +GC14DYMG$
$\Delta_f G(C14,Mg:Dy;0) = \underline{-GC14DYMG + 30000}$
$\Delta_f G(C14,Ho:Mg;0) = \underline{+GC14HOMG}$
$\Delta_f G(C14,Mg:Ho;0) = \underline{-GC14HOMG + 30000}$

C15: Model $(Al,R,Mg)_1(Al,R,Mg)_2$

Unary parameters
$\Delta_f G(C15,Al:Al;0) = +GLAVAL$
$\Delta_f G(C15,Gd:Gd;0) = GLAVGD$
$\Delta_f G(C15,Dy:Dy;0) = +GLAVDY$
$\Delta_f G(C15,Ho:Ho;0) = +GLAVHO$
$\Delta_f G(C15,Mg:Mg;0) = +GLAVMG$

Binary parameters
$\Delta_f G(C15,Gd:Al;0) = +GC15ALGD$
$\Delta_f G(C15,Al:Gd;0) = \underline{-GC15ALGD + 30000}$
$\Delta_f G(C15,Dy:Al;0) = +GC15ALDY$
$\Delta_f G(C15,Al:Dy;0) = \underline{-GC15ALDY + 30000}$
$\Delta_f G(C15,Ho:Al;0) = +GC15ALHO$
$\Delta_f G(C15,Al:Ho;0) = \underline{-GC15ALHO + 30000}$
$\Delta_f G(C15,Gd:Mg;0) = \underline{+GC15GDMG}$
$\Delta_f G(C15,Mg:Gd;0) = \underline{-GC15GDMG + 30000}$
$\Delta_f G(C15,Dy:Mg;0) = \underline{+GC15DYMG}$
$\Delta_f G(C15,Mg:Dy;0) = \underline{-GC15DYMG + 30000}$
$\Delta_f G(C15,Ho:Mg;0) = +GC15HOMG$
$\Delta_f G(C15,Mg:Ho;0) = \underline{-GC15HOMG + 30000}$
$\Delta_f G(C15,Mg:Al;0) = +15000$
$\Delta_f G(C15,Al:Mg;0) = +15000$

Ternary parameters
$L(C15,Gd:Al,Mg;0) = \underline{+25000}$

C36: Model $(Al,R,Mg)_1(Al,R,Mg)_2$

Unary parameters
$\Delta_f G(C36,Al:Al;0) = +GLAVAl$
$\Delta_f G(C36,Gd:Gd;0) = +GLAVGD$
$\Delta_f G(C36,Dy:Dy;0) = +GLAVDY$
$\Delta_f G(C36,Ho:Ho;0) = +GLAVHO$
$\Delta_f G(C36,Mg:Mg;0) = +GLAVMG$

Binary parameters
$\Delta_f G(C36,Gd:Al;0) = +GC36ALGD$
$\Delta_f G(C36,Al:Gd;0) = \underline{-GC36ALGD + 30000}$
$\Delta_f G(C36,Dy:Al;0) = +GC36ALDY$
$\Delta_f G(C36,Al:Dy;0) = \underline{-GC36ALDY + 30000}$
$\Delta_f G(C36,Ho:Al;0) = +GC36ALHO$
$\Delta_f G(C36,Al:Ho;0) = \underline{-GC36ALHO + 30000}$
$\Delta_f G(C36,Gd:Mg;0) = \underline{+GC36GDMG}$
$\Delta_f G(C36,Mg:Gd;0) = \underline{-GC36GDMG + 30000}$
$\Delta_f G(C36,Dy:Mg;0) = +GC36DYMG$
$\Delta_f G(C36,Mg:Dy;0) = \underline{-GC36DYMG + 30000}$
$\Delta_f G(C36,Ho:Mg;0) = +GC36HOMG$
$\Delta_f G(C36,Mg:Ho;0) = \underline{-GC36HOMG + 30000}$
$\Delta_f G(C36,Mg:Al;0) = \underline{-500}$
$\Delta_f G(C36,Al:Mg;0) = \underline{+30500}$

Ternary parameters
$L(C36,Gd,Mg:Al;0) = \underline{-45000 + 9\ T}$
$L(C36,Dy,Mg:Al;0) = \underline{-45000 + 7.5\ T}$
$L(C36,Ho,Mg:Al;0) = \underline{43000 + 5\ T}$

DYAL: Model $(R)_1(Al)_1$
$\Delta_f G(DYAL,Gd:Al;0) = \underline{-98000 + 23\ T}$
$\Delta_f G(DYAL,Dy:Al;0) = \underline{-99000 + 27.20\ T}$
$\Delta_f G(DYAL,Ho:Al;0) = \underline{-101000 + 20.87\ T}$

ZR3AL2: Model $(R)_3(Al)_2$
$\Delta_f G(ZR3AL2,Gd:Al;0) = \underline{-220000 + 55.89\ T}$
$\Delta_f G(ZR3AL2,Dy:Al;0) = \underline{-220000 + 64.78\ T}$
$\Delta_f G(ZR3AL2,Ho:Al;0) = \underline{-222500 + 48.00\ T}$

CO2SI: Model $(R)_2(Al)_1$
$\Delta_f G(CO2SI,Gd:Al;0) = \underline{-115500 + 30.40\ T}$
$\Delta_f G(CO2SI,Dy:Al;0) = \underline{-115,500 + 34.92\ T}$
$\Delta_f G(CO2SI,Ho:Al;0) = \underline{-120000 + 27.15\ T}$

A_NI3SN: Model $(Al)_3(Gd)_1$
$\Delta_f G(A_NI3SN,Al:Gd;0) = \underline{-165000 + 29.61\ T}$

DO24: Model $(Dy)_1(Al)_3$
$\Delta_f G(DO24,Dy:Al;0) = \underline{-170000 + 41.45\ T}$

HOAL3: Model $(R)_1(Al)_3$
$\Delta_f G(HOAL3,Dy:Al;0) = \underline{-150000 + 25.84\ T}$
$\Delta_f G(HOAL3,Ho:Al;0) = \underline{-172000 + 30.13\ T}$

BIF3: Model $(R)_1(Mg)_3$
$\Delta_f G(BIF3,Gd:Mg;0) = \underline{-50000 + 9.836\ T}$
$\Delta_f G(BIF3,Dy:Mg;0) = \underline{-49600 + 25.40\ T}$

GDMG5: Model $(Gd)_1(Mg)_5$
$\Delta_f G(GDMG5,Gd:Mg;0) = \underline{-52700 + 9.779\ T}$

ALMG_BETA: Model $(Mg)_{89}(Al)_{140}$
$\Delta_f G(ALMG_BETA,Mg:Al;0) = -246175 - 675.55\ T$

ALMG_EPS: Model $(Mg)_{23}(Al)_{30}$
$\Delta_f G(ALMG_EPS,Mg:Al;0) = -52565 - 173.18\ T$

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